Open Architectural Tour

I Built a Sovereign, Multi-Tenant CRM from Scratch.

A deep dive into building an enterprise sales engine with zero third-party cloud data lock-in, INR-first accounting, and sub-millisecond polymorphic querying.

p95 < 1ms	AES-256	100%
AST Query Latency	BYOK KMS Envelope	INR Lakhs / Crores

The Motivation

The Traditional CRM Trap: Bloat & Data Custody

Most enterprise CRMs hold tenant data hostage in proprietary silos, calculate currencies via rigid US-dollar assumptions, and slow teams down with sluggish interfaces.

💡 **Sovereignty Principle:** Organizations should own their encryption keys (BYOK), verify audit trails cryptographically via SHA-256 hash chains, and operate seamlessly in native ₹ INR units.

Core Engine

Polymorphic Metadata & Sub-ms Calculations

Dynamic schema authoring without SQL migration restarts. Define custom attributes, relationships, and formula ASTs that evaluate in microseconds.

4 Objects
Canonical Foundations

100% Dynamic
Custom Fields & Views

AST Lexer
Expression Engine

Sales Velocity

Visual Kanban with Deal Rotting Alerts

Smooth drag-and-drop opportunity tracking with automated stage rotting alerts, weighted revenue forecasting, and multi-currency exchange conversion.

⚡ **SLA-Driven Sales:** Automatically highlights deals stalled in stage for > 7 days and triggers escalation workflows before revenue leaks.

Automation

Zero-Effort Automated Activity Capture

Inbound email stitching via headers, telephony click-to-call, and SMS conversation logs stream directly into unified chronological timelines.

In-Reply-To
Thread Stitching

Auto Opt-Out
STOP / DNC Safety

Auto Unenroll
Drip Reply Safety

Data Integrity

4-Algorithm Fuzzy Identity Deduplication

Combines Jaro-Winkler, Levenshtein, Double Metaphone phonetic matching, and token-sort legal entity ratio to keep customer records pristine.

🛡️ **Survivorship Merge Engine:** Non-destructive field-level merging with cryptographic rollback snapshots.

Security

Cryptographically Verified Audit Trails & BYOK

Tenant data is enveloped with AES-256 data keys (DEK) wrapped by customer master keys (KEK). Every mutation is chained in a SHA-256 tamper-evident ledger.

SHA-256
Hash Chain Ledger

Zero Leakage
Tenant Isolation

1-Click
Audit Verification

Commerce

Quote-to-Cash in ₹ Lakhs/Crores & 18% GST

Built Indian-first. Formats all financial metrics according to Indian numbering standards (₹ 10,00,000) with line item CPQ approvals and GST invoicing.

🧾 **Compliance Out-of-the-Box:** 18% GST calculation, line-item discounts, and single-click invoice generation directly from closed deals.

User Experience

First-Run Magic: The 5-Stage Onboarding Flow

Zero sample data in production. Replaced with a thoughtful first-run journey: Welcome Screen → Feature Carousel → Coach Marks Tour → Checklist.

Bilingual	WCAG 2.2	0 Records
English & Hindi	AA Accessible	Clean Shipped App

Key Takeaway

Modern B2B Apps Can Be Fast, Sovereign & Beautiful.

Thank you for following along! Building Nexus CRM proved that high-performance enterprise capabilities and gorgeous design tokens belong together.

💬 Have you faced CRM data isolation or multi-currency issues in your team? Let's connect in the comments below!